Quantum Algorithms

Seminar Template

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Welcome

Welcome everyone! I'm so excited to have you in the course, and I hope these materials help you build both intuition and technical depth.

All lecture materials will be available on my personal homepage and on GitHub:

`https://github.com/red1108/lectures`

About Me

Hello, I am Mingyu Lee.

I am a fourth-year undergraduate from Seoul National University, currently an exchange student at UC Davis.

Personal homepage:
`https://red1108.github.io/`

LinkedIn:
`https://linkedin.com/in/red1108/`

Research Interests

I am studying quantum computation, quant trading, algorithms, and machine learning.

About This Course

Overview

- Major ideas in quantum algorithms
- Basic building blocks and frameworks
- Selected techniques with quantum complexity context
- Where quantum speedups may arise

Goals

- Build intuition for algorithms and complexity
- Learn several core quantum techniques
- Identify tasks that may benefit from quantum methods

About This Course

Plan

- Online format
- 5–10 participants
- Basic background, around Grover search
- 8 weeks

Notes

- Schedule can be adjusted to fit the group
- Detailed dates after participant recruitment
- Reading + discussion oriented progression

Tentative Schedule

Date	Study content	Person in charge	Additional notes
3/27 Fri	Quantum Complexity	Mingyu Lee	About oracle
4/3 Fri	Quantum Amplitude Amplification	Mingyu Lee	Grover search, quantum amplitude estimation
4/10 Fri	Quantum Min/Max Searching	Mingyu Lee	
4/17 Fri	Quantum Analog-Digital Conversion	Mingyu Lee	Swap test, k-NN
4/24 Fri	Quantum Walk	Mingyu Lee	
5/8 Fri	HHL	Mingyu Lee	
5/15 Fri	QSVT	Mingyu Lee	
5/22 Fri	Quadratic Speedup	Mingyu Lee	planted k-XOR, community detection

Outline

1. Introduction
2. Example Content
3. References

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- This is a starter Beamer deck for a new seminar.
- Theme and footer style follow your requested format settings.
- Contact information is placed on the title slide.

Outline

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Example: Key Topics

Topics in Quantum Algorithms

- Quantum Fourier Transform (QFT)
- Phase Estimation
- Grover Search
- Hamiltonian Simulation

Example: Simple Formula Slide

For example, the Hadamard transform on one qubit:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

This template is ready to be extended with your seminar materials. [NC10]

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References I

[NC10] Michael A Nielsen and Isaac L Chuang.
Quantum computation and quantum information.
Cambridge university press, 2010.